SMART DOCUMENT INTELLIGENCE SYSTEM

PROJECT REPORT

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1. ABSTRACT

The Smart Document Intelligence System is an AI-powered application that enables users to search, retrieve, and understand information from large collections of documents.

The system uses Retrieval-Augmented Generation (RAG) to provide accurate and context-aware answers.

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2. INTRODUCTION

Organizations store thousands of documents including reports, policies, manuals, and contracts.

Finding information manually is time-consuming.

This project aims to automate document search and question answering using Artificial Intelligence.

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3. OBJECTIVES

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1. Upload and process documents.
2. Convert text into vector embeddings.
3. Store embeddings in a vector database.
4. Retrieve relevant information for user queries.
5. Generate accurate responses using a Large Language Model.

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4. TECHNOLOGIES USED

Programming Language:

Python

Frontend:

Streamlit

Vector Database:

FAISS

Embedding Model:

all-MiniLM-L6-v2

Large Language Model:

Google Gemini

Libraries:

LangChain

Sentence Transformers

PyPDF

FAISS

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5. SYSTEM ARCHITECTURE

Document Upload

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Document Loader

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Text Splitter

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Embedding Generator

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FAISS Vector Database

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Retriever

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Gemini LLM

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Generated Answer

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6. IMPLEMENTATION

Step 1:

Load PDF documents.

Step 2:

Split documents into smaller chunks.

Step 3:

Generate embeddings using sentence-transformers.

Step 4:

Store embeddings in FAISS.

Step 5:

Retrieve relevant chunks based on user queries.

Step 6:

Generate final answers using Gemini.

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7. ADVANTAGES

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1. Faster information retrieval.
2. Improved employee productivity.
3. Reduced manual search effort.

4. Accurate document-based answers.

5. Scalable for large organizations.

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8. FUTURE ENHANCEMENTS

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1. Multi-language support.

2. Voice-based querying.

3. Real-time document updates.

4. Cloud deployment.

5. Enterprise authentication integration.

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9. CONCLUSION

The Smart Document Intelligence System demonstrates how Retrieval-Augmented Generation can transform traditional document management into an intelligent knowledge assistant.

The project significantly improves information accessibility and decision-making efficiency.